

Table 1: Clustering inertia on dataset collection 1 with 20.0% constraints (without musk-link constraints)

| Constraints Level<br>Method<br>Dataset | 0.2      |          |          |          |          |          |
|----------------------------------------|----------|----------|----------|----------|----------|----------|
|                                        | COPKM_P  | KSKM     | KSKM_E   | BLPKMCC  | PCCC     | COPKM    |
| appendicitis                           | 638.86   | 533.45   | 533.45   | 533.45   | 533.45   | -        |
| breast_cancer                          | 12214.56 | 12214.56 | 12214.56 | 12214.56 | 12214.56 | -        |
| bupa                                   | 2047.30  | 2047.30  | 2047.30  | 2047.30  | 2047.30  | -        |
| circles                                | 599.99   | 599.99   | 599.99   | 599.99   | 599.99   | -        |
| ecoli                                  | 1133.16  | 1059.03  | 966.81   | 1261.99  | 988.84   | 1615.28  |
| glass                                  | 1194.25  | 1116.94  | 1037.50  | 1143.20  | 1095.30  | -        |
| haberman                               | 893.98   | 890.90   | 890.90   | 890.90   | 890.94   | -        |
| hayes-roth                             | 608.25   | 601.79   | 535.76   | 546.98   | 536.08   | -        |
| heart                                  | 3120.52  | 3120.52  | 3120.52  | 3120.52  | 3120.52  | -        |
| ionosphere                             | 10971.49 | 10971.49 | 10971.49 | 10971.49 | 10971.49 | -        |
| led7digit                              | 2024.32  | 1769.52  | 1426.31  | 1772.80  | 1545.81  | -        |
| monk-2                                 | 2384.29  | 2384.29  | 2384.29  | 2384.29  | 2384.29  | -        |
| moons                                  | 322.93   | 322.93   | 322.93   | 322.93   | 322.93   | -        |
| movement_libras                        | 15155.20 | 13748.02 | 13096.30 | 13519.49 | 13163.94 | 15028.42 |
| newthyroid                             | 745.99   | 579.76   | 569.96   | 550.36   | 550.36   | 550.36   |
| saheart                                | 3927.71  | 3927.71  | 3927.71  | 3927.71  | 3927.71  | -        |
| sonar                                  | 11964.58 | 11947.37 | 11947.37 | 11947.37 | 11947.37 | -        |
| soybean                                | 367.14   | 367.14   | 367.14   | 414.92   | 367.14   | 380.53   |
| spectfheart                            | 11147.86 | 11147.86 | 11147.86 | 11147.86 | 11147.86 | -        |
| spiral                                 | 565.61   | 564.47   | 564.47   | 564.47   | 564.47   | -        |
| tae                                    | 731.36   | 719.54   | 681.81   | 683.70   | 678.54   | -        |
| vehicle                                | -        | -        | 13334.17 | 13334.17 | 13334.17 | -        |
| zoo                                    | 582.11   | 530.88   | 527.50   | 568.71   | 544.63   | 582.46   |